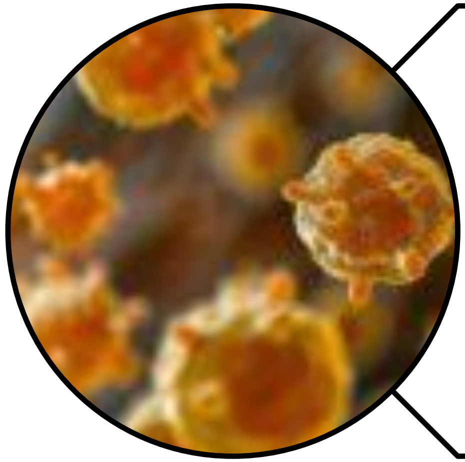
connecting people with science



Antimicrobial Resistance of Bacterial Pathogens – Africa

8-13 March 2026

NICD, Johannesburg, South Africa

Augusto Messa Jr.

CP1 – Introduction to Computational Methods

March 2026

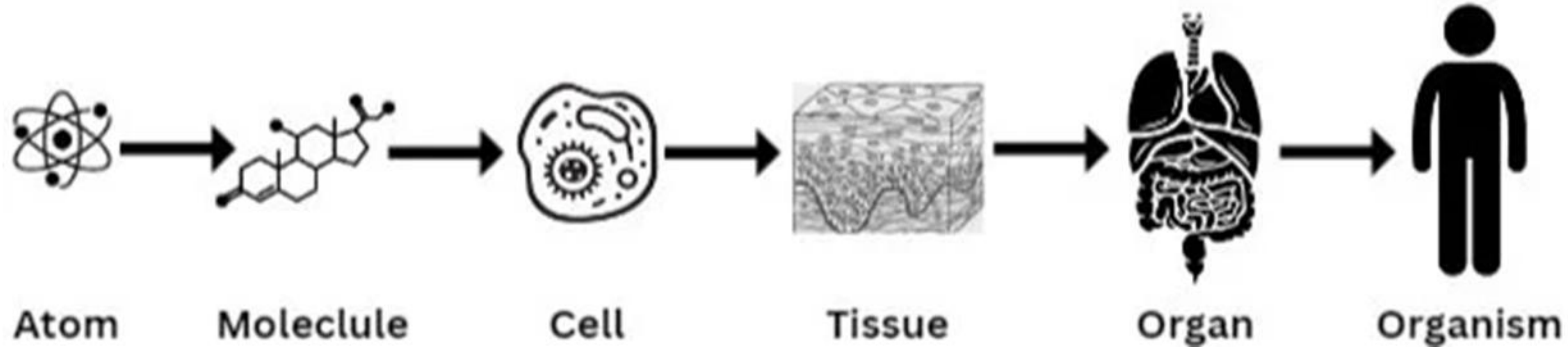
Learning objectives

- Provide a brief overview of Computational Methods for Biological Applications.
- Provide a brief overview of the workflow from sequences to AMR report.
- Explain some essential data formats: FASTA, FASTQ, SAM/BAM, VCF/BCF, GFF.

- Disclaimer: This is not (nor pretends to be) a comprehensive introduction to Computational Biology or Computational Methods for Biology, etc.

Computational biology

- Computational biology merges **computer science, mathematics (and statistics), and biology** to tackle complex biological puzzles.
    - **Using computers to analyze complex biological data** to gain insights on the biology of things: DNA sequences, protein structures.



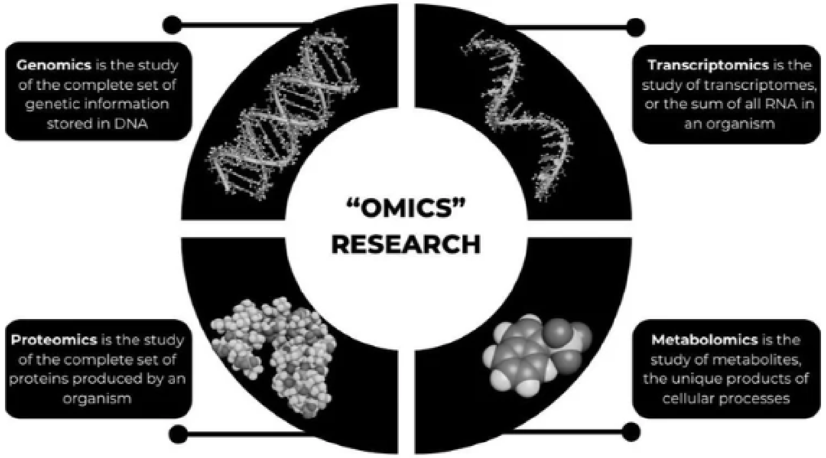
DNA, RNAs, proteins, lipids, carbohydrates, etc.

Computational biology - caveats

- **CB does not (and cannot) replace traditional biological lab research**; it complements it by helping translate large data into a testable hypothesis in labs.

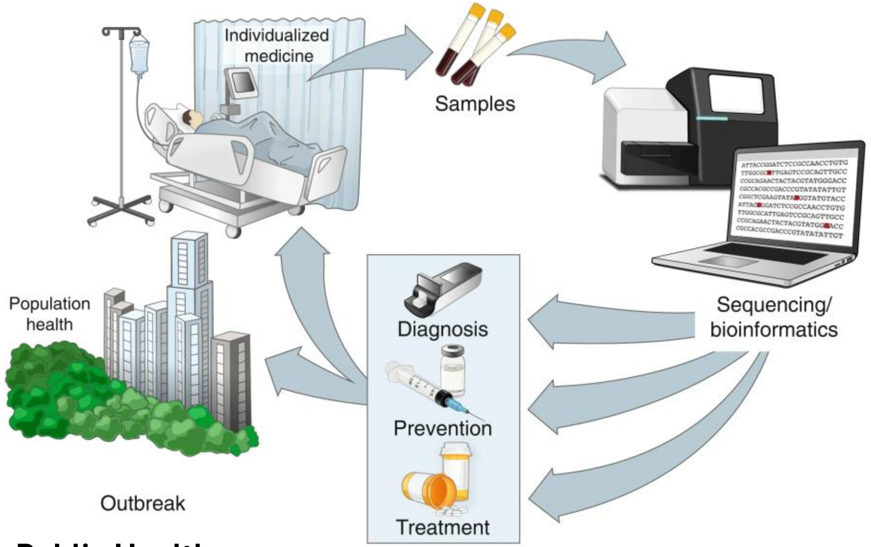
- **Golden rule of computational biology: garbage in, garbage out.**

- **Our results are only as good as our hypotheses, the quality of samples, and the quality of data.**



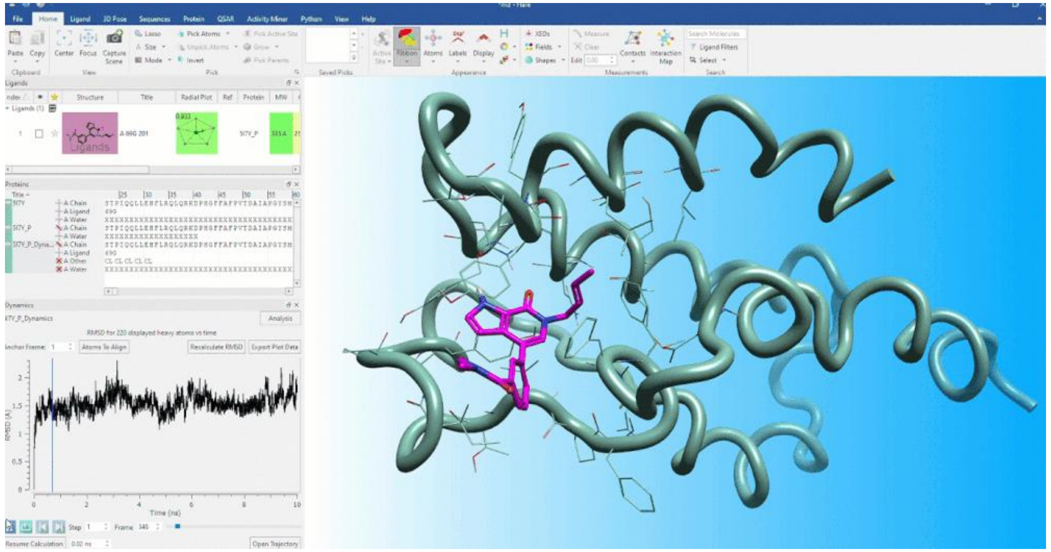
"Omics" Research

Simplify the analysis of large-scale biological data sets, which is generated by the growing interest in "omics" (genomics, transcriptomics, proteomics, metabolomics, etc.)



Public Health

Track pathogen evolution during outbreaks by analyzing sequenced samples to identify pathogens or variants and enable public health officials determine the most effective course of action. Ladner et al. 2019 (doi: 10.1038/s41591-019-0345-2)



Molecular dynamics simulations

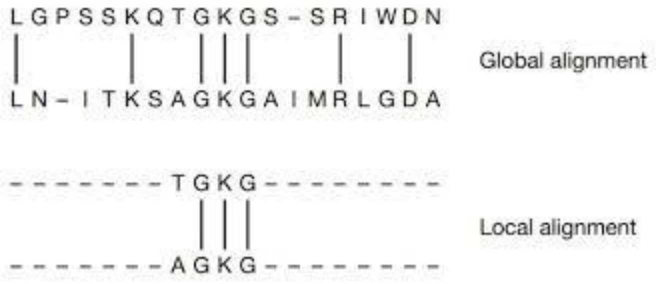
Provide detailed information about molecular conformations, interactions, and dynamics. Can be used for drug discovery, etc.

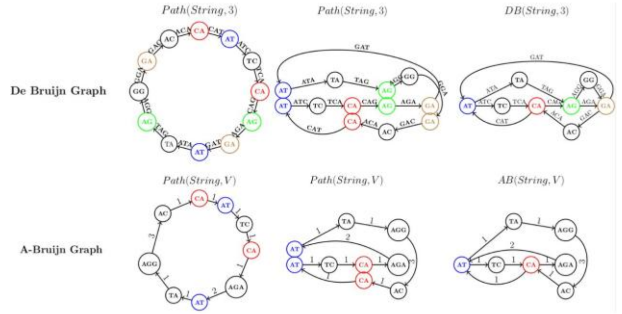
Pharmacogenomics

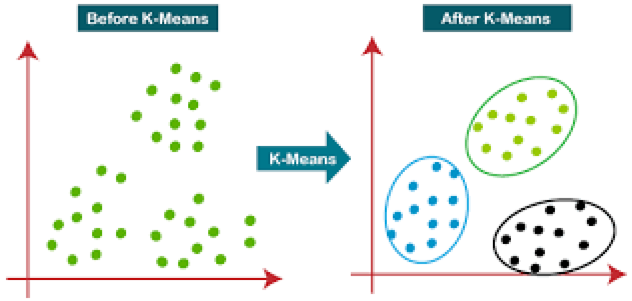
Provide insights into how certain genetic variants impact an individual's response to certain medications to help develop personalized treatment options.

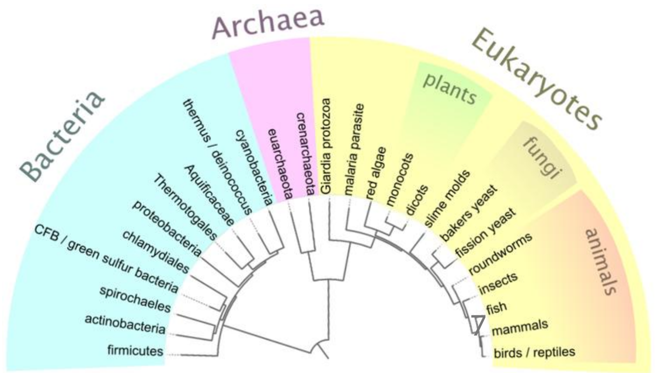
Basic algorithms, tools and techniques in CB

- **Sequence alignment algorithms:** arrange and compare sequences and identify regions of similarity (functional, structural or evolutionary relationships. E.g Needleman-Wunsch, Smith-Waterman, **BLAST**.
- **Sequence assembly algorithms**: reconstruct the complete sequence from short fragments (reads). E.g. de Bruijn graph-based methods.
- **Clustering algorithms**: group similar biological entities based on properties or measurements. E.g. hierarchical clustering, k-means.
- **ML algorithms**: build predictive models and identify patterns.
- **Phylogenetic inference algorithms**: reconstruct evolutionary relationships among organisms or sequences. E.g. Neighbor-Joining, Maximum-likelihood, Bayesian methods.

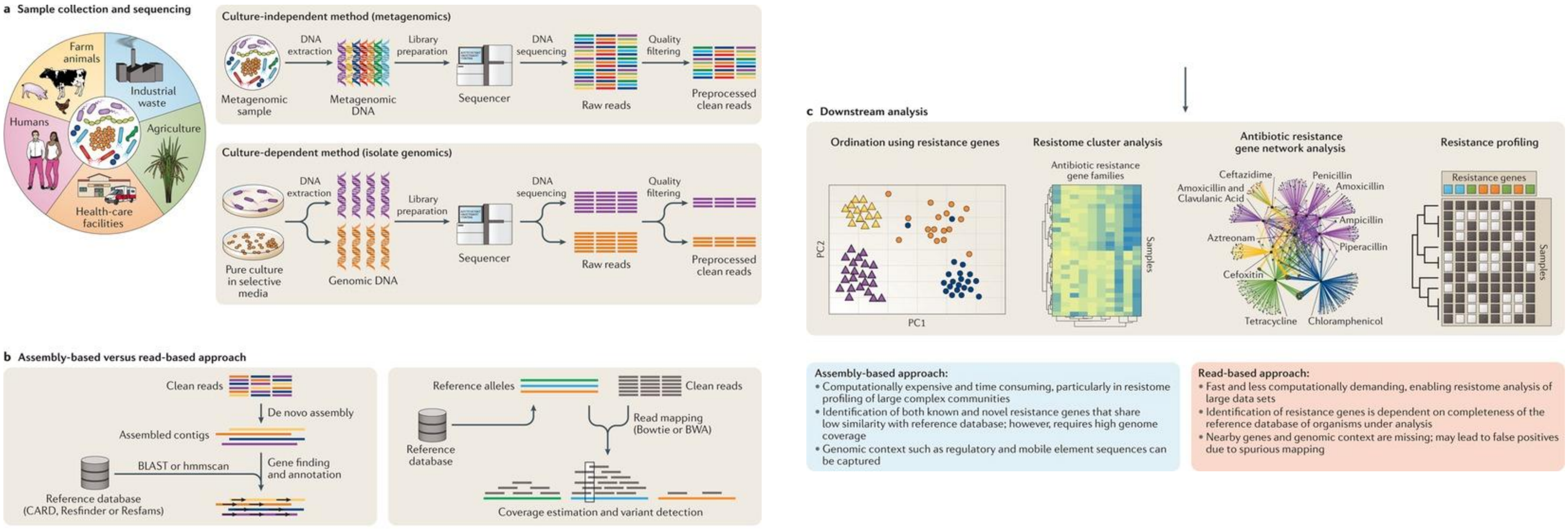






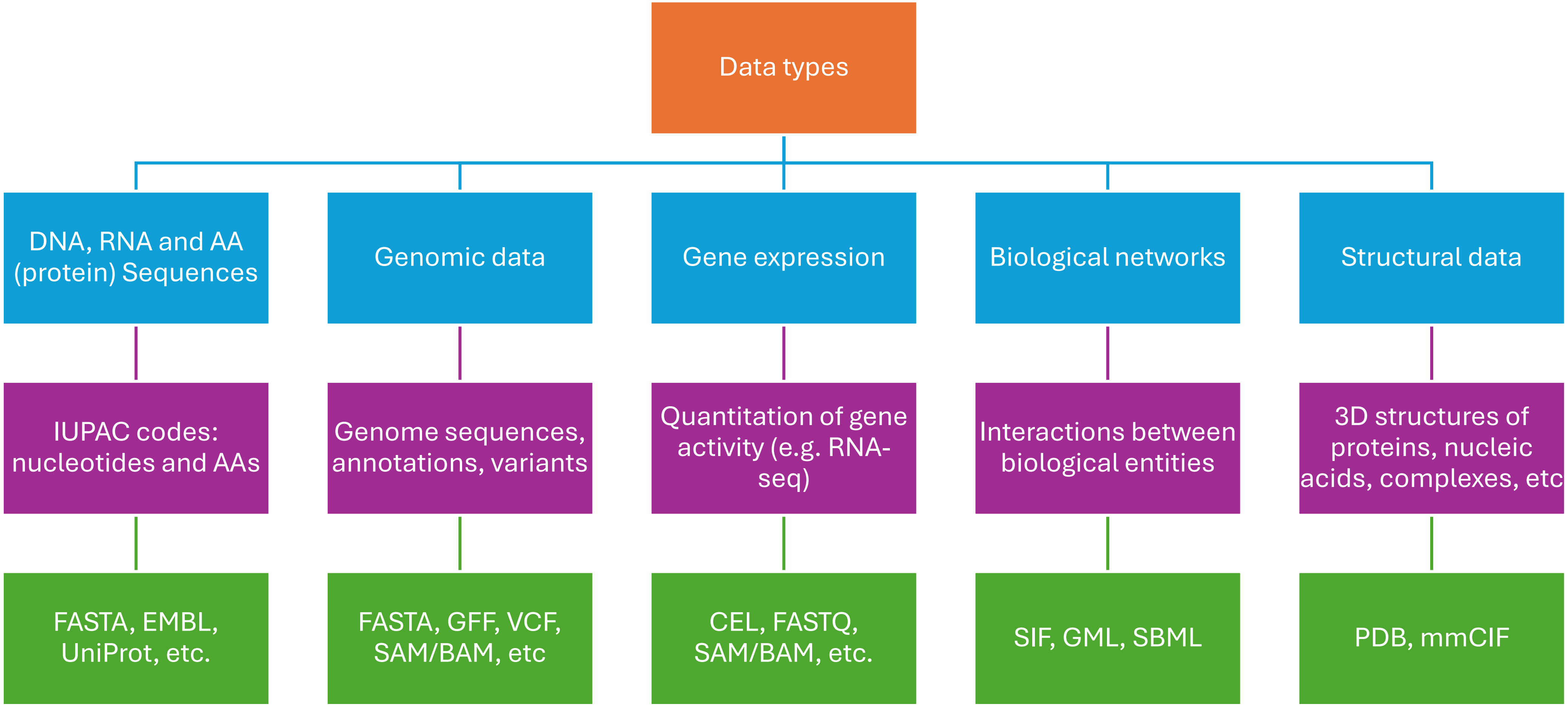


Assembly- vs. read-mapping-based approaches to study AMR using genomic technologies



Adapted from: Boolchandani, D'Souza and Dantas. Sequencing-based methods and resources to study AMR. [10.1038/s41576-019-0108-4](https://doi.org/10.1038/s41576-019-0108-4)

(Some) Biological data types and formats



PS: This is NOT, by far, a comprehensive list of what is out there.
PS2: This is not a formal hierarchy, it is just to help you comprehend how it works.

Sequence formats (FASTA and FASTQ)

- Both are ubiquitous text-based formats to represent sequences. They usually contain one or multiple sequences.
- Both begin with a label/ID for each sequence, followed by the sequence itself. FASTQ extends FASTA.
- Can be compressed.

```
>gi|1817694395|ref|NZ_JAAGMU010000151.1| Streptomyces sp. SID7958 contig-52000002, whole genome shotgun sequen
CCGGCTGGCGCGGCTGGCGCTGGCGGTGGGGCTGCGGCTGCTGGAGCTGGGGGTGGCGCTGGAGGCGCAC
GGCCAGAACCTGCTGGTGGTGCTGTCGCCGTCCGGGGAGCCGCGGCGGCTGGTCTACCGCGATCTGGCGG
ACATCCGGGTCTCCCCCGCGCGGCTGGCCCGGCACGGTATCCGGGTTCCGGACCTGCCGGCG

>gi|1643051563|gb|SZWM01000399.1| Citrobacter sp. TBCS-14 contig3128, whole genome shotgun sequence
GCACAGTGAGATCAGCATTCCGTTGGATCTACTGGTCAATCAAAACCTGACGCTGGGTACTGAATGGAAC
CAGCAGCGCATGAAGGACATGCTGTCTAACTCGCAGACCTTTATGGGCGGTAATATTCCAGGCTACAGCA
GCACCGATCGCAGCCCATATTCGAAAGCCGAGATCTTCTCTTTGTTTGCCGAAAACAACATG
```

FASTA is a common format for assembled genomes.
Extensions: “.fasta”, “.fna”, “.faa”, “.frn”

FASTQ is the *de facto* standard for data from NGS and 3rd generation sequencing technologies.
We will refer to them very often, also as “reads”.

ASCII table and corresponding Q-scores

Symbol	ASCII Code	Q-Score	Symbol	ASCII Code	Q-Score
!	33	0	6	54	21
"	34	1	7	55	22
#	35	2	8	56	23
$	36	3	9	57	24
%	37	4	:	58	25
&	38	5	;	59	26
'	39	6	<	60	27
(	40	7	=	61	28
)	41	8	>	62	29
*	42	9	?	63	30
+	43	10	@	64	31
,	44	11	A	65	32
-	45	12	B	66	33
.	46	13	C	67	34
/	47	14	D	68	35
0	48	15	E	69	36
1	49	16	F	70	37
2	50	17	G	71	38
3	51	18	H	72	39
4	52	19	I	73	40
5	53	20			

Q scores represent the sequencer's estimate of the probability that the nucleotide at the position was called incorrectly.

$$Q=-10\log_{10}p$$

p is the probability that the position was called incorrectly.

In this sense if:

Q=10 (ASCII: +), error probability is 1 in 10 (10%=0.1)

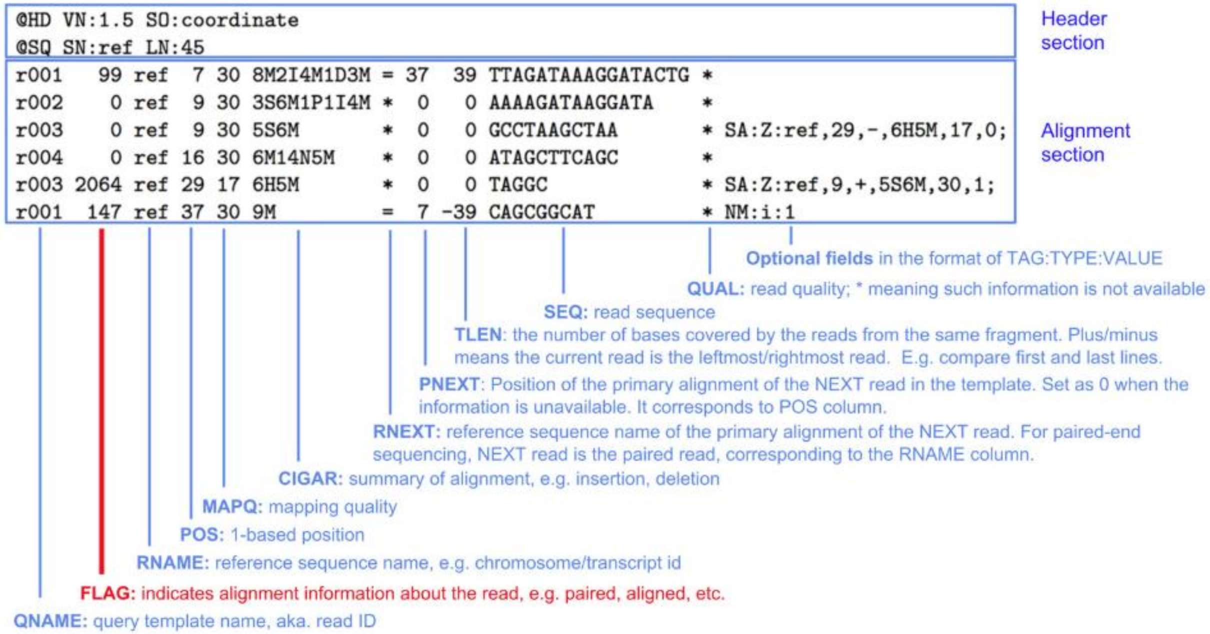
Q=20 (ASCII: 5), error probability is 1 in 100 (1%=0.01)

Q=30 (ASCII: ?), error probability is 1 in 1000 (0.1%=0.001).

Don't focus on memorizing symbols, but rather their meaning.

Alignment formats (SAM/BAM/CRAM)

- These represent how a sequence is aligned relative to a reference sequence (i.e. involves coordinates).

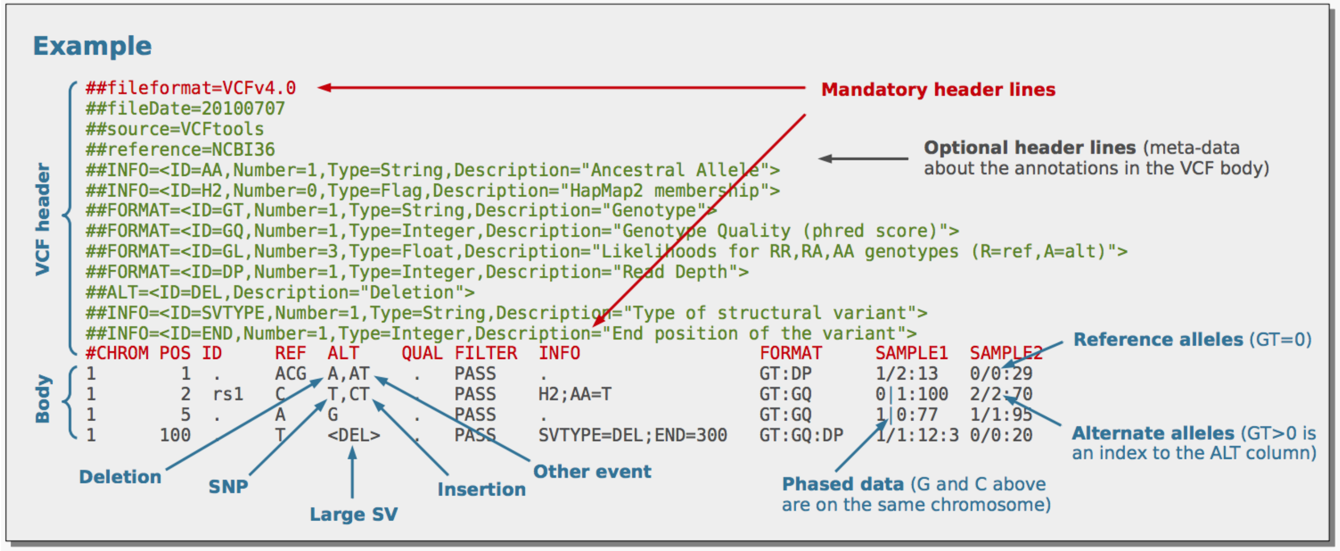


Binary Alignment Map (BAM) is a binary representation of SAM, containing the same information in binary format for **improved performance**.

CRAM is a sequencing read file format that is highly space efficient by using reference-based compression of sequence data. Files are typically 30 to 60% **smaller** than their BAM equivalents

Variant information formats

- The Variant Call Format (VCF) is a format for storing variations (e.g., polymorphisms) between a reference genome and sequences aligned to it, based on SAM/BAM alignments



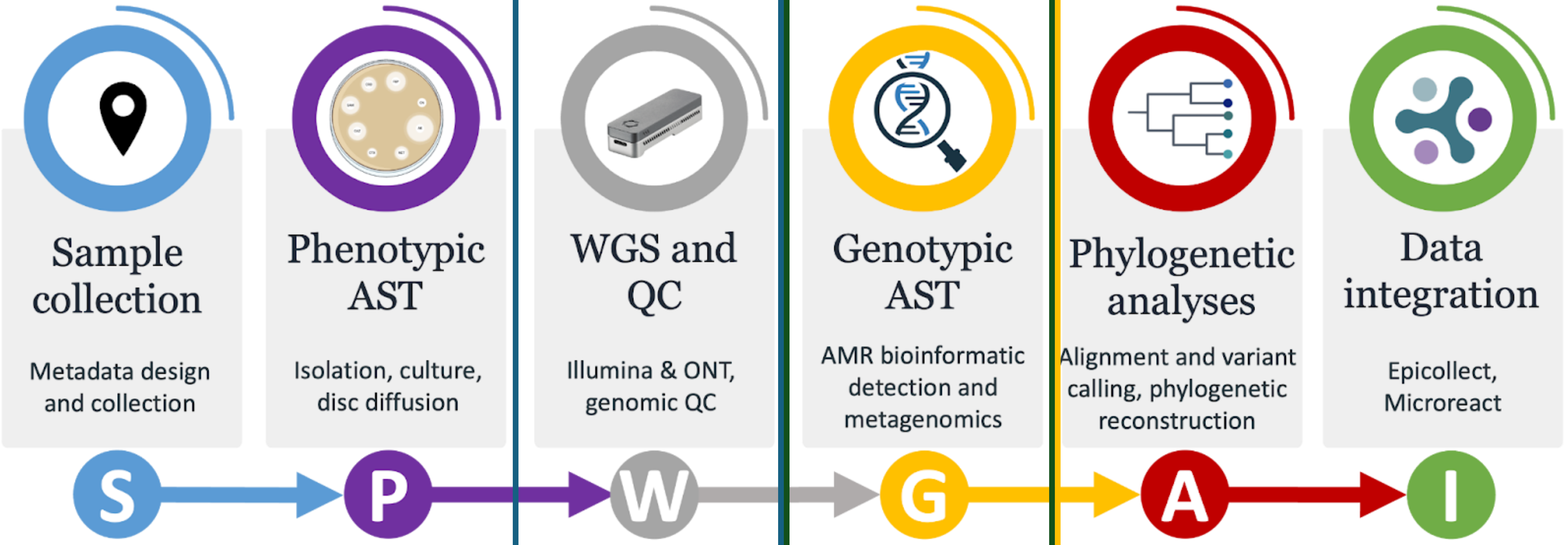
Binary Call Format (BCF) is a binary representation of VCF, containing the same information in binary format for **improved performance**.

Annotation formats

- Annotation: is the process of identifying and labelling/tagging the regions of the genome that correspond to functional elements.
- Feature: is the term used to refer to these functional elements that were identified through annotation and giving them the coordinates, names and more metadata.

```
##gff-version 3.2.1
##sequence-region ctg123 1 1497228
ctg123 . gene            1000  9000  .  +  .  ID=gene00001;Name=EDEN
ctg123 . TF_binding_site 1000  1012  .  +  .  ID=tfbs00001;Parent=gene00001
ctg123 . mRNA            1050  9000  .  +  .  ID=mRNA00001;Parent=gene00001;Name=EDEN.1
ctg123 . mRNA            1050  9000  .  +  .  ID=mRNA00002;Parent=gene00001;Name=EDEN.2
ctg123 . mRNA            1300  9000  .  +  .  ID=mRNA00003;Parent=gene00001;Name=EDEN.3
ctg123 . exon            1300  1500  .  +  .  ID=exon00001;Parent=mRNA00003
ctg123 . exon            1050  1500  .  +  .  ID=exon00002;Parent=mRNA00001,mRNA00002
ctg123 . exon            3000  3902  .  +  .  ID=exon00003;Parent=mRNA00001,mRNA00003
ctg123 . exon            5000  5500  .  +  .  ID=exon00004;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . exon            7000  9000  .  +  .  ID=exon00005;Parent=mRNA00001,mRNA00002,mRNA00003
ctg123 . CDS             1201  1500  .  +  0  ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
```

Workflow from sample to AMR report

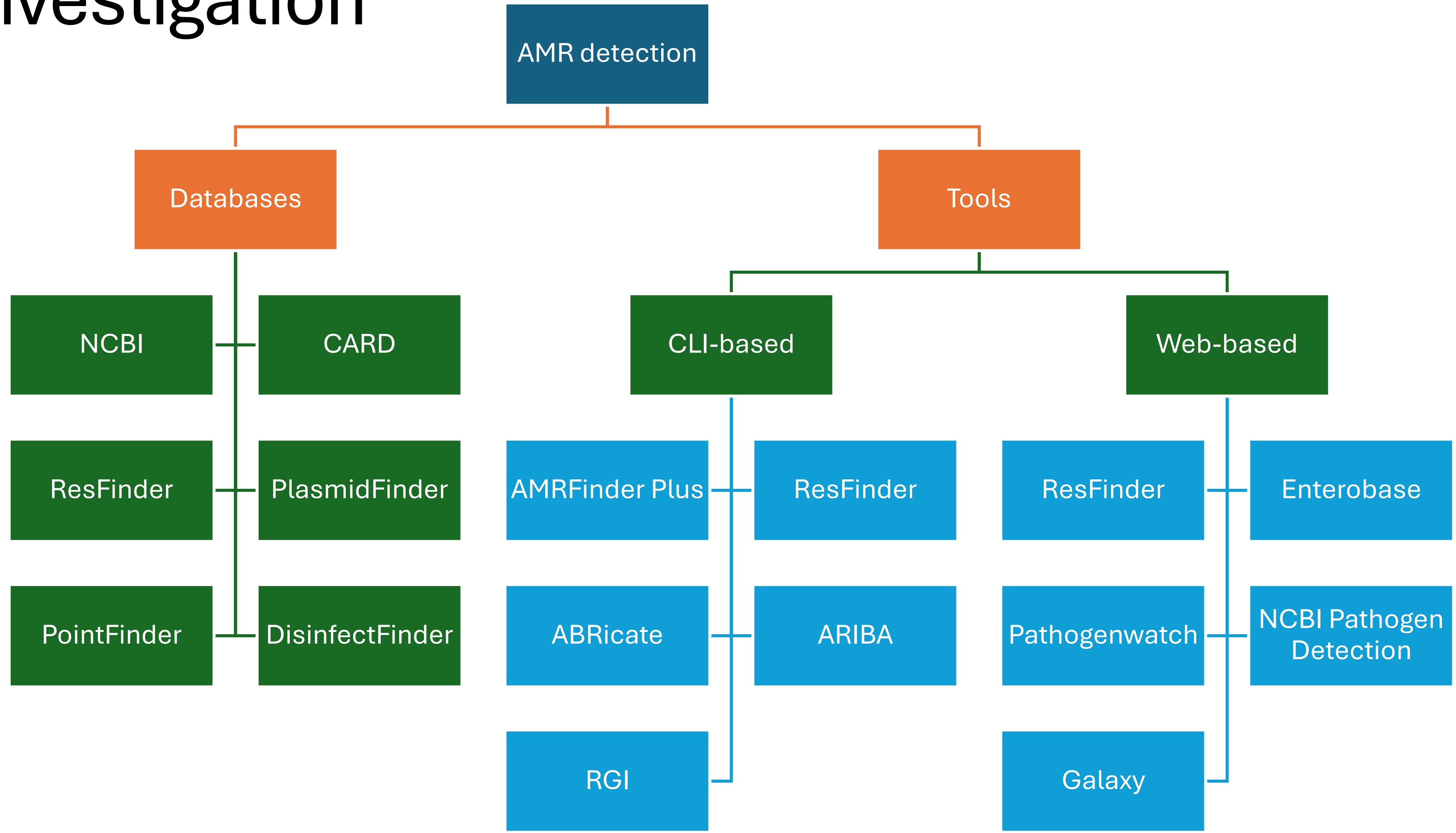


FASTQ (reads) → FASTA (assembled)

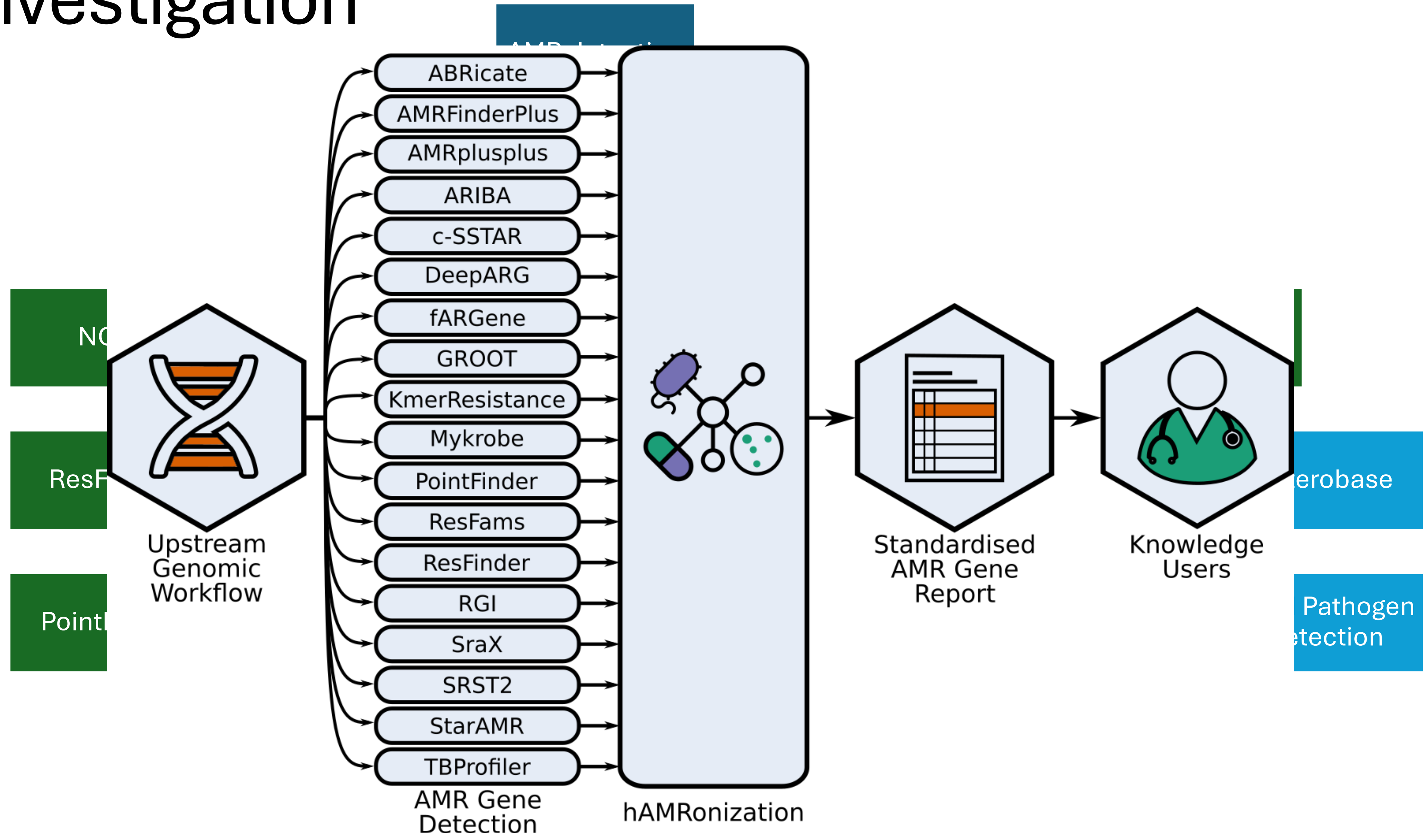
FASTA (reference DBs): NCBI, ResFinder, CARD, other

SAM/BAM, VCF, GFF; Newick (Phylogenetic trees)

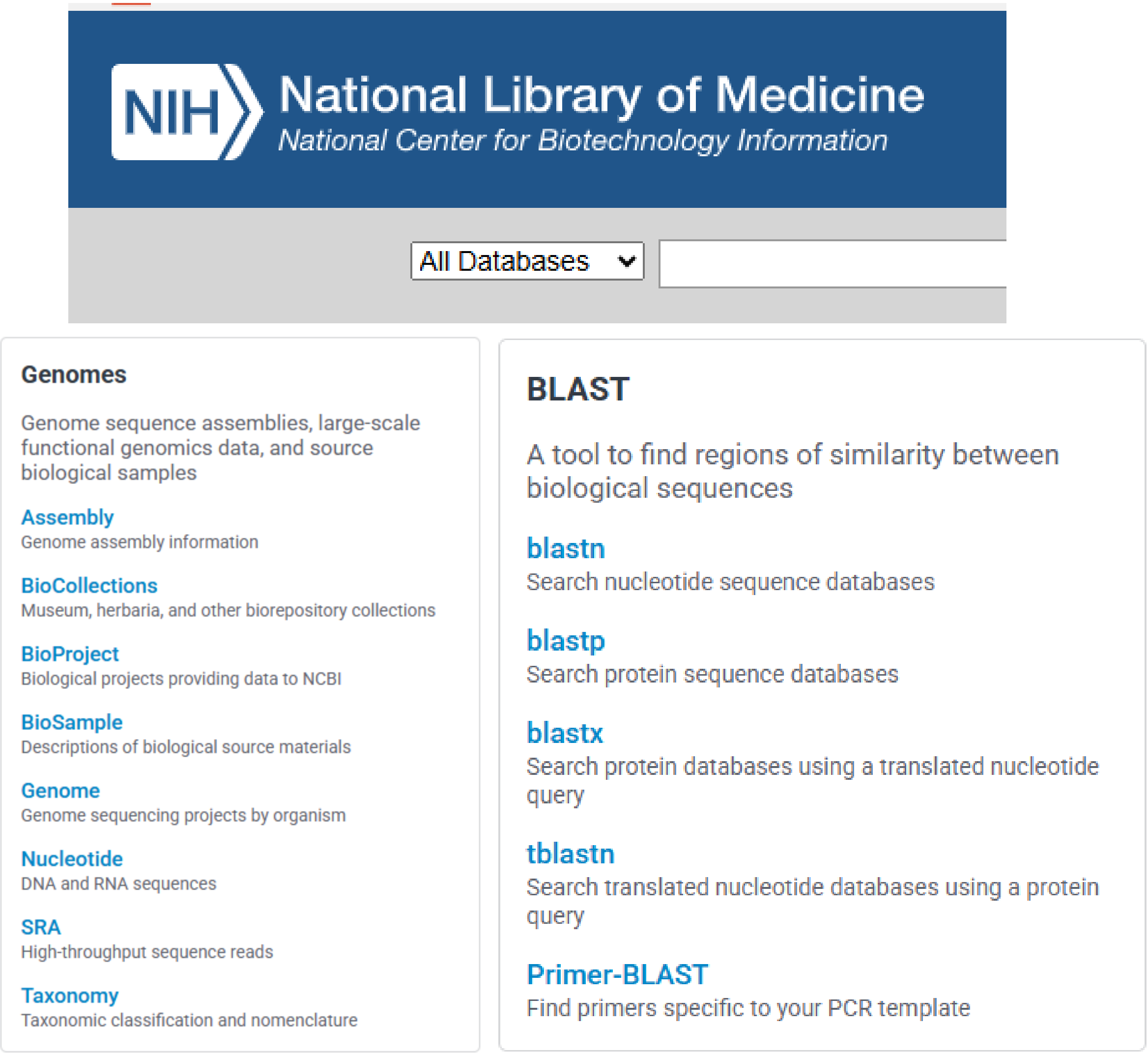
Computational Methods and DBs for AMR investigation



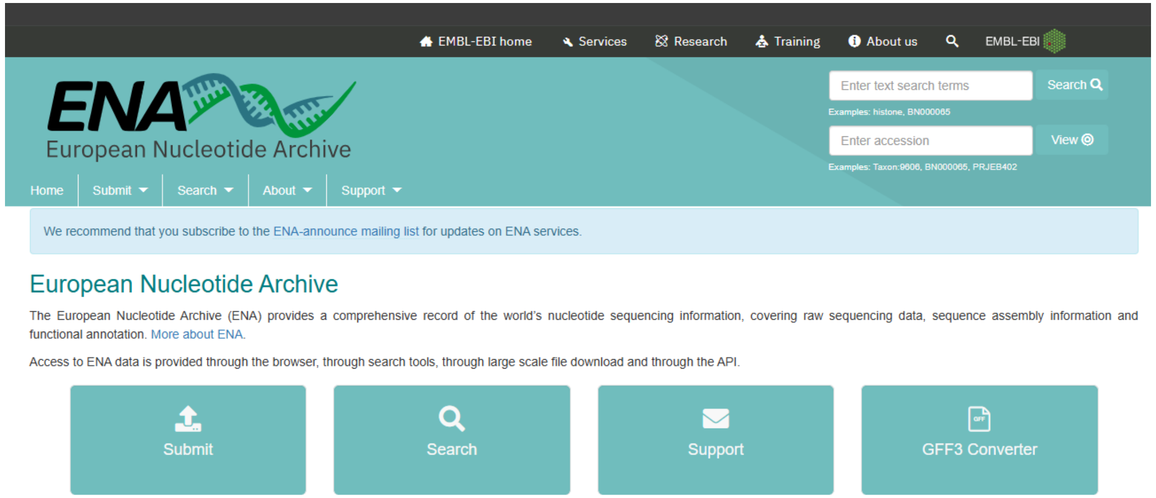
Computational Methods and DBs for AMR investigation



Sequence Databases (used often)



ncbi.nlm.nih.gov/search/



ebi.ac.uk/ena/browser/home

Suggested readings

- Boolchandani, M., D'Souza, A.W. & Dantas, G. Sequencing-based methods and resources to study antimicrobial resistance. Nat Rev Genet 20, 356–370 (2019). Doi: 10.1038/s41576-019-0108-4
- Konopka, A.K., Crabbe, M.J. Introduction: Computational Biology in 14 Brief Paragraphs. In Konopka, A.K., Crabbe, M.J, editors. Compact Handbook of Computational Biology. CRC Press. 2004, 6 p.
- https://fiveable.me/computational-biology/unit-1
- https://www.biosparked.com/course/niche-eszfp-zw9hf-ygfzr
- https://rnnh.github.io/bioinfo-notebook/docs/file_formats.html
- https://samformat.pages.dev/sam-format-flag
- https://support.nlm.nih.gov/kbArticle/?pn=KA-03574

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